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Arch Linux on WSL2 Setup Guide

This guide explains how to run the Windows script that installs Arch Linux in WSL2 on Windows 11 and adds internet speed test tools such as `iperf3` and `speedtest-cli`.

What this does

The script turns on Windows Subsystem for Linux (WSL), a way to run Linux on Windows. It sets WSL2 as the default version, downloads the Arch Linux WSL image, imports it as a WSL distribution, and installs the speed test tools inside Arch Linux.

Before you start

- Use a Windows 11 computer with administrator access.[web:3]
- Make sure virtualization is enabled in the computer BIOS or UEFI if WSL installation fails.
- Save the script as a file ending in `.cmd`, such as `install-arch-wsl-speedtest.cmd`.
- Keep the computer connected to the internet during setup.

Step 1: Save the script (Unnecessary if you use the wsl.cmd file in this folder)

1. Open **Notepad**.
2. Paste the script into the blank file.
3. Select **File** and then **Save As**.
4. In **File name**, enter `awsl.cmd`.
5. In **Save as type**, choose **All Files**.
6. Save it somewhere easy to find, such as the **Desktop**.

Step 2: Run the script

1. Find the `.cmd` file you saved.
2. Right-click it.
3. Select **Run as administrator**.
4. Wait while Windows installs WSL, configures WSL2, downloads Arch Linux, and installs packages.
5. If Windows asks to restart, restart the computer and run the script again.

Step 3: Open Arch Linux

After the script finishes, open **Command Prompt**, **PowerShell**, or **Windows Terminal** and enter:

```
wsl -d archlinux
```

This starts the Arch Linux environment that was imported by the script.[web:5][web:11]

Step 4: Sign in

The script creates this user account unless it already exists:

- Username: `fieldtech`
- Password: `fieldtech`

After signing in, it is a good idea to change the password.

To change the password, enter:

```
passwd
```

Step 5: Run a speed test

To run the installed speed test tools, use these commands inside Arch Linux:

```
speedtest-cli
```

```
iperf3 -c <server> -t 30 -P 4
```

`speedtest-cli` checks internet speed using a command line speed test tool, and `iperf3` tests throughput against an iperf server.[web:16][web:5]

Example:

```
iperf3 -c 192.168.1.10 -t 30 -P 4
```

What to expect

- The first run can take several minutes.[web:3][web:5]
- WSL may need a restart before settings fully apply.[web:3]
- Arch Linux will open in a terminal window, not a desktop environment.[web:5]
- The script writes a `.wslconfig` file in the Windows user profile to tune WSL2 resource settings.[web:3]

If something goes wrong

Try these steps:

1. Restart the computer.
2. Run the script again as administrator.
3. Open Command Prompt as administrator and run:

```
wsl --status
```

4. If WSL is missing or broken, run:

```
wsl --update
```

5. If installation still fails, check that virtualization is enabled in BIOS or UEFI and that Windows features for WSL and Virtual Machine Platform are available.[web:3][web:12]

Useful commands

Open Arch Linux:

```
wsl -d archlinux
```

See installed WSL distributions:

```
wsl -l -v
```

Shut down WSL:

```
wsl --shutdown
```

Run a speed test:

```
speedtest-cli
```

Run an iperf test:

```
iperf3 -c <server> -t 30 -P 4
```